Towards a Science of Human-AI Decision Making: An Overview of Design Space in Empirical Human-Subject Studies

FAccT 2023


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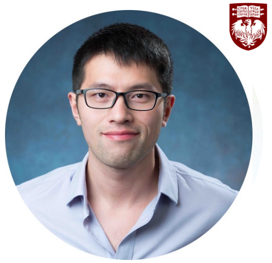
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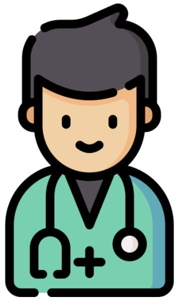
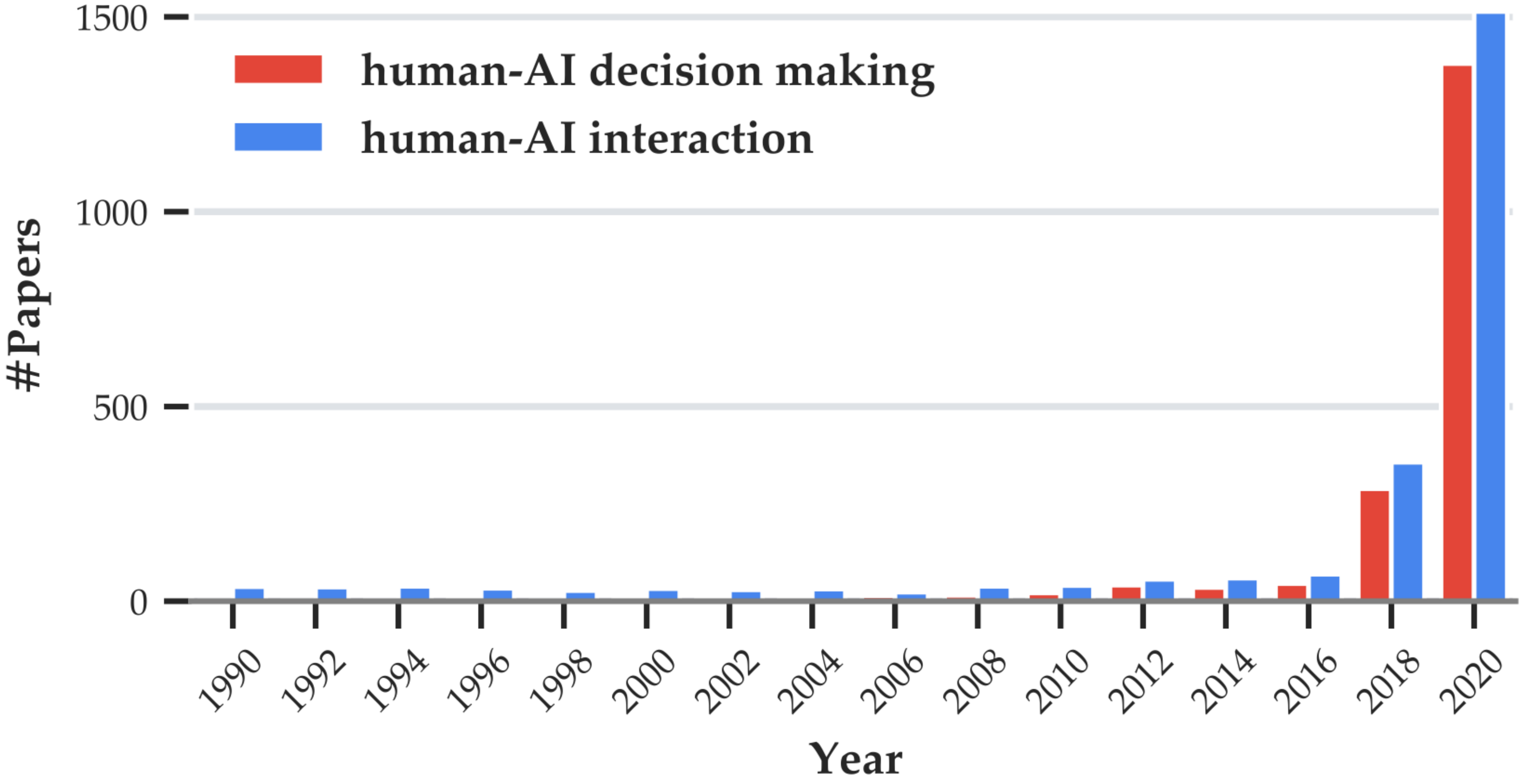
Promise of AI in improving human-AI decision making

Recidivism prediction

Medical diagnosis



Loan approval



1500
1000
500
0
#Papers
human-AI decision making
human-AI interaction
1990
1992
1994
1996
1998
2000
2002
2004
2006
2008
2010
2012
2014
2016
2018
2020
Year

? *Human + AI < AI*

Looking into the future

Looking into the future

*1. How should we design **human-subject studies**?*

Looking into the future

1. *How should we design* ***human-subject studies****?*

2. *What are the current* ***trends****?*

Looking into the future

1. *How should we design* ***human-subject studies****?*

2. *What are the current* ***trends****?*

3. *What are the* ***gaps and recommendations*** *for the field?*

What we did ✍️

100+ papers

Three aspects:

1. Decision tasks
2. AI models & AI assistance elements
3. Evaluation metrics

Three aspects:

1. Decision tasks
2. AI models & AI assistance elements
3. Evaluation metrics

Current trends

Current gaps

Future Recommendations

Recidivism prediction

Medical diagnosis

Loan approval

1. Decision tasks

Domain	Decision task
Law & Civic	Recidivism prediction [58, 94, 139, 143] and its slight variations [39, 115], likelihood to recidivate [54, 55, 94], bail outcomes prediction [7, 61, 82], child maltreatment risk prediction [35].
Medicine & Healthcare	Medical disease diagnosis [82], cancer image search [24], cancer image classification [73], COVID-19 diagnosis [137], balance disorder diagnosis [22], clinical notes annotation/medical coding [90], stroke rehabilitation assessment [86, 87]
Finance & Business	Income prediction [62, 115, 144, 155], loan approval [15, 55, 139], loan risk prediction [29], sales forecast [36, 97], property price prediction [1, 111], apartment price prediction [101] selecting overbooked airline passengers for re-routing [15], determining to freeze bank accounts due to money laundering suspicion [15], stock price prediction [16], marketing email prediction [60], dynamically pricing car insurance premiums [15].
Education	Students' performance forecasting [36, 37, 144], student admission prediction [7, 28], student dropout prediction [82], LSAT question answering [13].
Leisure	Music recommendation [76, 77], movie recommendation [78], song rank order prediction [95], speed dating [96, 152], Facebook news feed prioritization [112], Quizbowl [43], draw-and-guess [23] word guessing [48], chess playing [32], plant classification [147], goods division [88].
Professional	Job promotion [15], meeting scheduling assistance [74], email topic classification [125], cybersecurity monitoring [42], profession prediction [94], military planning (monitor and direct unmanned vehicles) [130].

Current trends

1. Variety

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Law & Civic	Recidivism prediction [58, 94, 139, 143] and its s… to recidivate [54, 55, 94], bail outcomes predictio… prediction [35].
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Professional	Job promotion [15], meeting scheduling assistanc… cybersecurity monitoring [42], profession predictio… direct unmanned vehicles) [130].
Artificial	Alien medicine recommendation [79, 104], alien re… bean counting [109], broken glass prediction [153],… reading time prediction [134], water pipe failure p…
Generic	Question answering [27, 52], image classification… 106].
Others	Deception detection [80, 81, 94], forest cover predi… nutrition prediction [20, 21], person weight estim…

Current trends

1. Variety
2. Task characteristics, mostly **high-stake** tasks.

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Current gaps

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1. Choice of tasks are driven by datasets availability

Current trends

Current gaps

1. Variety
2. Task characteristics, mostly high-stake tasks.

1. Choice of tasks are driven by datasets availability
2. Lack of frameworks for generalizable knowledge

Current trends

1. Variety
2. Task characteristics, mostly high-stake tasks.

Current gaps

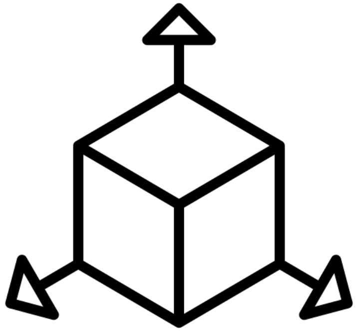
1. Choice of tasks are driven by datasets availability
2. Lack of frameworks for generalizable knowledge
3. Misalignment with application reality

Future Recommendations

1. Develop frameworks to characterize the space of decision tasks

Consider 4 dims:

1. Risk
2. Required expertise
3. Subjectivity
4. Emulation vs. discovery



Future Recommendations

1. Develop frameworks to characterize the space of decision tasks

2. Justify choices of decision task

Rationales

Future Recommendations

1. Develop frameworks to characterize the space of **decision tasks**

2. Justify choices of decision task

3. Expand datasets availability

2. AI models & AI assistance

AI models

Model	Examples
Deep learning models	Convolution Neural Networks [3, 27], Recurrent Neural Networks [26, 60], BERT [80], RoBERTa [13], VQA model (hybrid LSTM and CNN) [115], not specified [23, 24, 30, 48, 50, 52, 62, 73, 86, 87, 107, 110, 152]
"Shallow" models	Logistic regression [16, 39, 41, 45, 68, 86, 106, 143], linear regression [28, 37, 101, 111], generalized additive models* (GAMs) [1, 13, 39, 43, 49, 128, 135] decision trees/random forests [29, 45, 54, 55, 86, 92, 97, 137, 144, 155], support-vector machines (SVMs) [41, 80, 81, 86, 94, 114, 147, 152], Bayesian decision lists [82], K-nearest neighbors [77], shallow (1- to 2-layer) neural networks [45, 106], naive Bayes [125], matrix factorization [78]
Wizard of Oz	[7, 15, 20–22, 79, 95, 96, 104, 109, 139]

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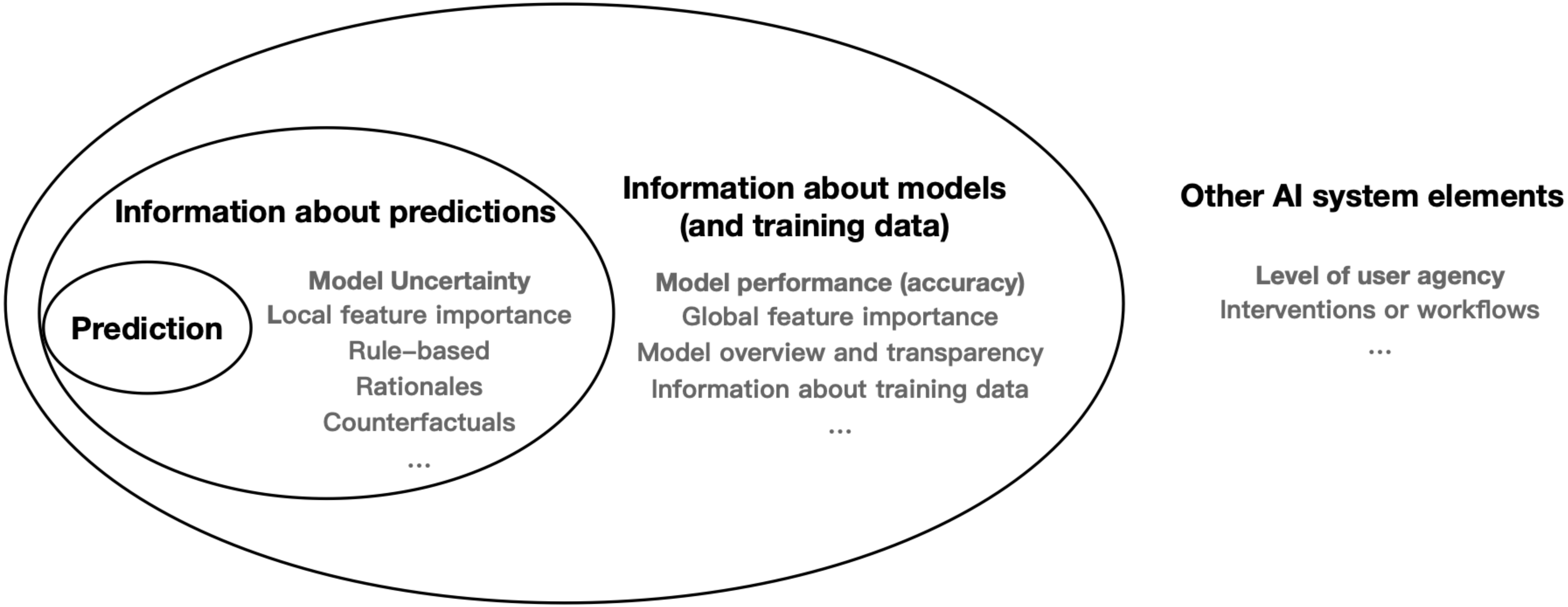
Taxonomy AI assistance elements



Taxonomy AI assistance elements

Information about predictions

Prediction

Model Uncertainty
Local feature importance
Rule–based
Rationales
Counterfactuals
...

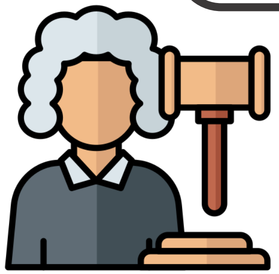
Information about models (and training data)

Model performance (accuracy)
Global feature importance
Model overview and transparency
Information about training data
...

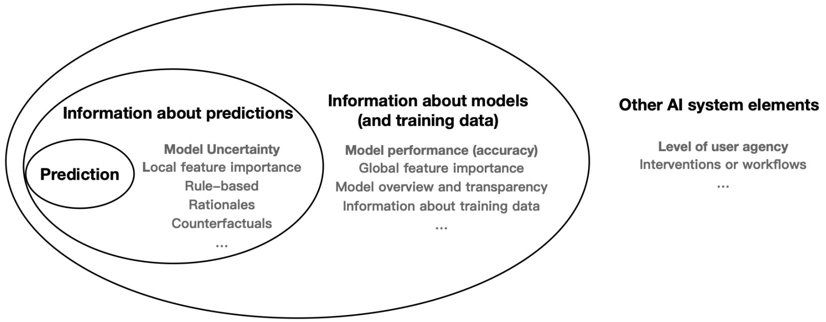
Other AI system elements

Level of user agency
Interventions or workflows
...

Recidivism prediction



Recidivism prediction



Prediction:

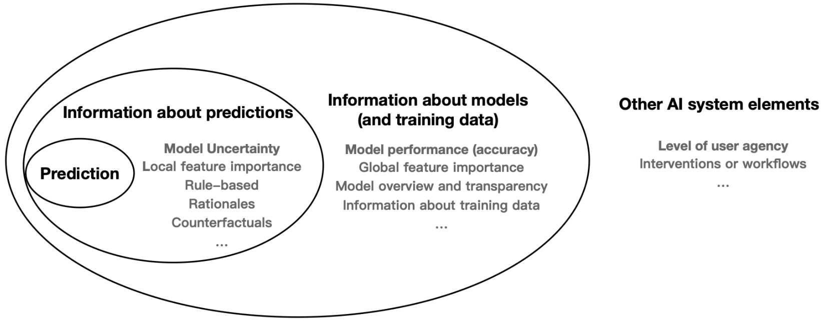
- Positive/negative

AI Assistance Elements	Methods
	Model predictions
Prediction	Binary prediction [12, 13, 15, 16, 28, 39, 58, 62, 80, 81, 92, 94, 96, 107, 125, 137, 143, 152, 153, 155], multi-class [13, 21–23, 32, 42, 43, 52, 60, 76–78, 88, 90, 94, 109, 130, 147], continuous prediction or regression [1, 10, 29, 35–37, 54, 55, 73, 86, 87, 95, 109, 111, 128, 134], multiple predictions or multi-step decision [21, 73, 79, 86–88, 90, 104], with alternates [10, 13, 42, 43, 60, 130]
No prediction shown	[3, 7, 11, 12, 24, 27, 45, 48, 61, 74, 82, 97, 106, 112, 139, 144]

Recidivism prediction

Information about prediction:

- 80% confidence
- Important features: # prior crimes; crime degree

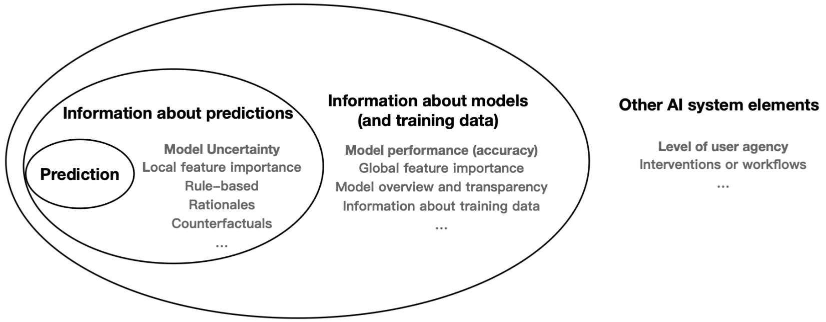


AI Assistance Elements	Methods
	Information about model predictions
Model uncertainty	Classification confidence (or probability) [10, 13, 21, 22, 43, 60, 68, 73, 86, 87, 144, 155], flag low confidence [90], uncertainty distribution (regression) [101]
Local feature importance	Coefficients [16, 28, 39, 43, 49, 55, 80, 81, 86, 87, 94, 106, 111, 125, 128, 143], attention [26, 27, 80], gradient-based [27, 73, 106], propagation-based (LRP [3]), perturbation-based (LIME [3, 13, 62, 106], SHAP [29, 144, 155]), Wizard of Oz [15, 20–22], video features [107]
Rule-based explanations	Decision sets [79, 82, 104], tree-based explanation [77, 92], anchors [62, 115]
Example-based methods	Nearest neighbor or similar training instances [15, 20, 23, 24, 39, 62, 77, 78, 81, 137, 143]
Counterfactual explanations	Contrastive or sensitive features [39, 97], counterfactual examples [15, 45, 92, 143]
Natural language explanations	Model-generated rationales [16, 32, 137], expert-generated rationales [13]
Partial decision boundary	Traversing the latent space around a data input [62]

Recidivism prediction

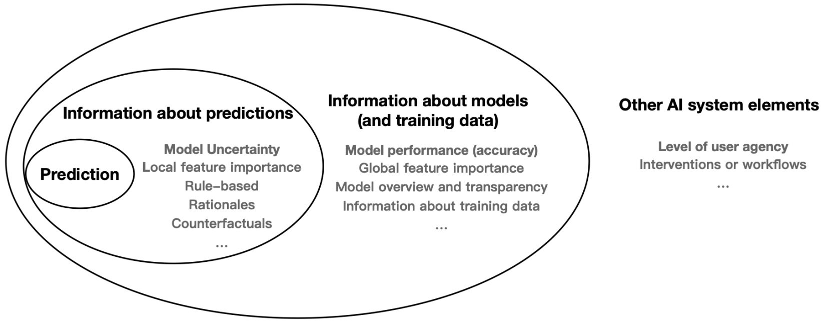
Information about the model

- Model accuracy: 65% accuracy



AI Assistance Elements	Methods
	Information about models (and training data)
Model performance	Accuracy [61, 80, 81, 147, 152], false positive rates [61]
Global feature importance	Coefficients [39], permutation-based [143], shape function of GAMs [1], Wizard of Oz [15]
Presentation of simple models	Decision sets [82], decision trees [45], linear regression [111], logistic regression [45], one-layer MLP [45]
Global example-based explanations	Model tutorial [80], prototypes [23, 43, 74]
Model documentation	Overview of the model or algorithm [74, 76, 77, 88, 112], model prediction distribution [139]
Information about training data	Input features or information the model considers [61, 77, 111, 155], aggregate statistics (e.g., demographic) [15, 39], full training “data explanation” [7]

Recidivism prediction



Other AI elements:

- Level of user agency: incorporate user feedback

AI Assistance Elements	Methods
Other AI system elements affecting user agency or experience	
Interventions or workflows affecting cognitive process	User makes prediction before model [21, 58, 96, 111, 143, 152, 155], vary system response times [21, 109], outcome feedback to user [12, 13, 27, 58, 147, 153], training phase [27, 76, 80, 111, 155], source of recommendation or local explanation (human or AI) [36, 78, 95], varied model quality [42, 74, 90, 107, 109, 125, 152, 153]
Levels of user agency	Allowing user feedback or personalization for model [43, 76, 125], outcome control (after decision [37, 88, 155], before decision [74]), interactive explanations [24, 28, 94], user direction on input data [24, 73, 90], level of machine agency [21, 90]

AI assistance elements

Recidivism prediction

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- Positive/negative

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- Level of user agency: incorporate user feedback

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Table 3. AI assistance elements explored in prior work on human-AI decision making, grouped into four categories: (1) model predictions, (2) information about model predictions, (3) information about models (and training data), and (4) other AI system elements affecting user agency or experience.

Current trends

1. Assistance beyond predictions

Current trends

1. Assistance beyond predictions
2. A focus on AI explanations

Current trends

1. Assistance beyond predictions
2. A focus on AI explanations
3. Beyond the model

Current trends

Current gaps

1. Assistance beyond predictions
2. A focus on AI explanations
3. Beyond the model

1. No systematic understanding and limited design space of AI assistance elements

Current trends

Current gaps

1. Assistance beyond predictions
2. A focus on AI explanations
3. Beyond the model

1. No systematic understanding and limited design space of AI assistance elements
2. Focus on decision trials only instead of the holistic experience with AI

Current trends

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1. Assistance beyond predictions
2. A focus on AI explanations
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Current gaps

1. No systematic understanding and limited design space of AI assistance elements
2. Focus on decision trials only instead of the holistic experience with AI
3. Generalizability

Future Recommendations

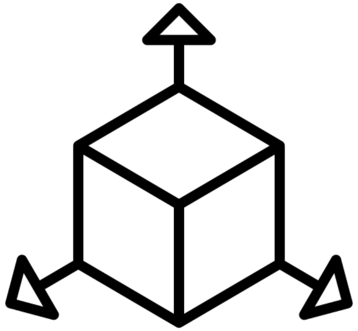
1. Human-centered analysis to define the design space of AI assistance elements

A top-down frameworks

Human-centered
instead of technical availability

Future Recommendations

1. Human-centered analysis to define the design space of AI assistance elements



A top-down frameworks

Human-centered
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Information about prediction:

- 80% confidence
- Important features: # prior crimes; crime degree

Future Recommendations

1. Human-centered analysis to define the design space of AI assistance elements

2. Extend the design space and studies beyond **decision trials**

Decision trial:

Seeing an instance and making a decision with AI's assistance

Fail to account for

- Individual difference
- Prior experience
- On-boarding experience
- Workload
- Etc.…

Future Recommendations

1. Human-centered analysis to define the design space of AI assistance elements

2. Extend the design space and studies beyond decision trials

3. Understand user needs and behaviors for a certain **task**

Decision tasks are chosen in an **ad-hoc fashion** or even **arbitrarily**.

More based on the **availability of current resources** (explanations methods or datasets)

3. Evaluation metrics

Metrics w.r.t. tasks

Evaluation	Subjective?	Metrics
Efficacy	Subjective	Self-rated error/accuracy [37, 81, 137], perceived performance improvement [32], confidence in the decisions [54, 55, 60, 95], soundness of participants' mental models [76]
	Objective	Accuracy/error [11, 13, 16, 20, 20, 24, 26, 42, 43, 52, 58, 62, 68, 73, 74, 79–81, 90, 92, 94, 97, 99, 101, 107, 130, 144, 155], F1 [16, 104], precision [16], recall [16, 90], AUC-ROC [41], false positive rate [26, 54, 99], false negative rate [26, 41], true positive rate [41], true negative rate [41], Brier score [54, 55], mean prediction error [111], win rate [32, 48], cumulative award [52], customized score/return [12], human percentile rank [32], agreement between labels [87]
Efficiency	Objective	time taken on the task (response time, average time for a game round, speed) [1, 10, 26, 28, 45, 48, 52, 74, 79, 82, 90, 92, 104, 125, 130, 144, 147], total number of labels [90]
Task satisfaction & mental demand	Subjective	Satisfaction with the process [37], confidence in the process [37], frustration/annoyance [77, 125], mental demand/effort [20, 21, 77, 144], workload [24, 87, 128, 130], task difficulty [10]
	Objective	number of words in user feedback [82]

Metrics w.r.t. AI

Evaluation	Subjective?	Metrics
Understanding	Subjective	Self-reported understanding [7, 15, 20, 23, 28, 97, 125, 143, 147], confidence in understanding [76], confidence in simulation [3, 106], ease of understanding [60, 111], intuitiveness [134], perceived transparency/interpretability [112, 137]
	Objective	Forward simulation [1, 3, 20, 27, 29, 45, 62, 106, 107, 111, 115, 143, 153], counterfactual simulation [62, 143], model errors detection [143], identifying important features [28, 134], correctness of described model behaviors [29, 77, 112], correctness of estimated model performance/accuracy [36, 55, 107, 125], comprehension quiz [28, 48, 76, 143]
Trust and reliance	Subjective	Self-reported trust [1, 3, 20, 28, 29, 36, 45, 48, 55, 76, 111, 111, 115, 125, 128, 137], model confidence/acceptance [3, 29, 77, 125, 134, 143], self-reported agreement/reliance [27], perceived accuracy [74, 125, 128], perceived capability/benevolence/integrity [107, 112], usage intention/willingness [1, 24, 29, 36, 74, 77, 106, 112]
	Objective	Agreement/acceptance of model suggestions [13, 16, 22, 26, 35, 80, 81, 90, 94, 96, 101, 143, 152, 153, 155], switch [58, 96, 101, 109, 152, 155], weight of advice [95, 111], model influence (difference between conditions) [54, 55], disagreement/deviation [111], choice to use the model [11, 36, 37, 114], over-reliance [21, 22, 143, 147], under-reliance [22, 143, 147], appropriate reliance [52, 111, 143, 147]
Fairness	Subjective	Perceived fairness [7, 39, 54, 61, 139], individual fairness [88], group fairness [88], process fairness [15], deserved outcome [15], feature fairness [15, 139], accountability [112]
	Objective	Decision bias [54, 55]
System satisfaction and usability	Subjective	Satisfaction [16, 37, 74, 79, 97, 104, 137], helpfulness/support [16, 20, 24, 42, 74, 147], usefulness [60, 86], effectiveness [137], quality [137], appropriateness [20], preference/likability [76, 86, 115, 147], system affect [128], system usability [130], complexity [21], ease/comfort of use [7, 137], system frustration [74, 86, 125], richness/informativeness [7, 86, 87], learning [137], recommendation to others [74]
	Objective	time spent on the application [23]
Others	Subjective	Rate on specific features (e.g., explanations): quality/soundness/completeness of explanation [77, 78, 134], usefulness/helpfulness of explanation [24, 87, 107, 134], agreement with explanation [144], easiness to use explanation [134], explanation workload [1], attribution to AI versus self [23], desire to provide feedback [125], expected model improvement [125]

Current trends

1. Diverse evaluation metrics

Current trends

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2. A focus on efficacy when evaluating decision tasks

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	Objective	number of words in user feedback [82]

Current trends

1. Diverse evaluation focuses
2. A focus on efficacy when evaluating decision tasks
3. Focuses on understanding, trust, system satisfaction, and fairness w.r.t. AI

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	Objective	Forward simulation [1, 3, 20, 27, 29, 45, 62, 106, 107, 111, 115, 143, 153], terfactual simulation [62, 143], model errors detection [143], identifying i tant features [28, 134], correctness of described model behaviors [29, 77, correctness of estimated model performance/accuracy [36, 55, 107, 125], prehension quiz [28, 48, 76, 143]
Trust and reliance	Subjective	Self-reported trust [1, 3, 20, 28, 29, 36, 45, 48, 55, 76, 111, 111, 115, 125, 128, model confidence/acceptance [3, 29, 77, 125, 134, 143], self-reported a ment/reliance [27], perceived accuracy [74, 125, 128], perceived ca ity/benevolence/integrity [107, 112], usage intention/willingness [1, 24, 2 74, 77, 106, 112]
	Objective	Agreement/acceptance of model suggestions [13, 16, 22, 26, 35, 80, 8 94, 96, 101, 143, 152, 153, 155], switch [58, 96, 101, 109, 152, 155], weig advice [95, 111], model influence (difference between conditions) [54 disagreement/deviation [111], choice to use the model [11, 36, 37, over-reliance [21, 22, 143, 147], under-reliance [22, 143, 147], approp reliance [52, 111, 143, 147]
Fairness	Subjective	Perceived fairness [7, 39, 54, 61, 139], individual fairness [88], group ness [88], process fairness [15], deserved outcome [15], feature fairnes 139], accountability [112]
	Objective	Decision bias [54, 55]
System satisfaction and usability	Subjective	Satisfaction [16, 37, 74, 79, 97, 104, 137], helpfulness/support [16, 20, 2 74, 147], usefulness [60, 86], effectiveness [137], quality [137], approp ness [20], preference/likability [76, 86, 115, 147], system affect [128], sy usability [130], complexity [21], ease/comfort of use [7, 137], system fr tion [74, 86, 125], richness/informativeness [7, 86, 87], learning [137], re mendation to others [74]
	Objective	time spent on the application [23]
Others	Subjective	Rate on specific features (e.g., explanations): quality/soundness/complet of explanation [77, 78, 134], usefulness/helpfulness of explanation [24, 87 134], agreement with explanation [144], easiness to use explanation explanation workload [1], attribution to AI versus self [23], desire to pr feedback [125], expected model improvement [125]

Current trends

1. Diverse evaluation metrics
2. A focus on efficacy when evaluating decision tasks
3. Focuses on understanding, trust, system satisfaction, and fairness w.r.t. AI
4. A lack of **consensus** on measurements

Evaluation	Subjective?	Metrics
Understanding	Subjective	Self-reported understanding [7, 15, 20, 23, 28, 97, 125, 143, 147], confiden in understanding [76], confidence in simulation [3, 106], ease of understan ing [60, 111], intuitiveness [134], perceived transparency/interpretability [11 137]
	Objective	Forward simulation [1, 3, 20, 27, 29, 45, 62, 106, 107, 111, 115, 143, 153], cou terfactual simulation [62, 143], model errors detection [143], identifying impo tant features [28, 134], correctness of described model behaviors [29, 77, 11 correctness of estimated model performance/accuracy [36, 55, 107, 125], co prehension quiz [28, 48, 76, 143]
Trust and reliance	Subjective	Self-reported trust [1, 3, 20, 28, 29, 36, 45, 48, 55, 76, 111, 111, 115, 125, 128, 13 model confidence/acceptance [3, 29, 77, 125, 134, 143], self-reported agre ment/reliance [27], perceived accuracy [74, 125, 128], perceived capab ity/benevolence/integrity [107, 112], usage intention/willingness [1, 24, 29, 3 74, 77, 106, 112]
	Objective	Agreement/acceptance of model suggestions [13, 16, 22, 26, 35, 80, 81, 9 94, 96, 101, 143, 152, 153, 155], switch [58, 96, 101, 109, 152, 155], weight advice [95, 111], model influence (difference between conditions) [54, 5 disagreement/deviation [111], choice to use the model [11, 36, 37, 11 over-reliance [21, 22, 143, 147], under-reliance [22, 143, 147], appropria reliance [52, 111, 143, 147]
Fairness	Subjective	Perceived fairness [7, 39, 54, 61, 139], individual fairness [88], group fa ness [88], process fairness [15], deserved outcome [15], feature fairness [1 139], accountability [112]
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	Objective	time spent on the application [23]
Others	Subjective	Rate on specific features (e.g., explanations): quality/soundness/completene of explanation [77, 78, 134], usefulness/helpfulness of explanation [24, 87, 1 134], agreement with explanation [144], easiness to use explanation [13 explanation workload [1], attribution to AI versus self [23], desire to provi feedback [125], expected model improvement [125]

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Current gaps

1. A focus on decision efficacy (rather than human experience)

Current trends

Current gaps

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2. A focus on efficacy when evaluating decision tasks
3. Focuses on understanding, trust, system satisfaction, and fairness w.r.t. AI
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1. A focus on decision efficacy (rather than human experience)
2. Use of subjective vs. objective measurements need to be better understood and regulated

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Current gaps

1. A focus on decision efficacy (rather than human experience)
2. Use of subjective vs. objective measurements need to be better understood and regulated
3. Subjective measures are difficult to replicate
4. **Variance** on the coverage of measurements; lack of a framework

Future Recommendations

1. Validate choices of evaluation metrics by research questions/hypotheses and targeted constructs

Measurement validity from statistics and social science

- Construct validity
- Content validity

Future Recommendations

1. Make choices of evaluation metrics by research questions/hypotheses and targeted constructs

2. Work towards common metrics

Key to rigorous and replicable science

Future Recommendations

1. Make choices of evaluation metrics by research questions/hypotheses and targeted constructs

2. Work towards common metrics

Evaluation metrics can profoundly shape the outcomes of a field.

"Measure what is measurable, and make measurable what is not so."

- Galileo Galilei

Last but not least!

Survey website 👉

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Informational Notes

This interactive table allows you to search papers by keywords in various columns. For example, you are able to find papers with "explanations" in
click on the Reset button.

Reset

There are currently **81** papers in the table.

Paper	Paper link	Authors	Venue	Year	Task	AI m	
Human Decision Making with Machine Assistance: An Experiment on Bailing and Jailing	Link	Nina Grgic-Hlaca, Christoph Engel, Krishna P. Gummadi	CSCW	2019	Recidivism prediction		
Progressive Disclosure: Designing for Effective Transparency	Link	Aaron Springer, Steve Whittaker	IUI	2019	Emotion analysis	Generalized additive models (GAMs)	Prediction (continuous p regression)
COGAM: Measuring and Moderating Cognitive Load in Machine Learning Model Explanations	Link	Ashraf Abdul, Christian von der Weth, Mohan Kankanhalli, Brian Y. Lim	CHI	2020	Property price prediction	Generalized additive models (GAMs)	• Prediction (continuou prediction or regressi • Global feature import (shape function of G/

https://haidecisionmaking.github.io

Thank you

Survey website 👉

https://haidecisionmaking.github.io

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